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# Self-Refined Distillation

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Anonymous Author(s)

Affiliation

Address

email

## Abstract

1 On-policy distillation (OPD) has become a central post-training tool for large  
2 language models (LLMs), providing dense per-token teacher supervision along the  
3 student’s own rollouts. Its parameter-sharing variant, on-policy self-distillation  
4 (OPSD), uses the same model as both student and teacher, with the teacher addition-  
5 ally conditioned on privileged information such as the ground-truth answer. Despite  
6 its promise, OPSD suffers from failure modes that existing remedies address only  
7 through token-level loss interventions, such as truncation or reweighting. In this  
8 work, we identify a common structural cause underlying several such failures,  
9 which we call *prefix failure*. Under prefix failure, dense per-token supervision  
10 induces a bimodal teacher mixture and fragmented gradients that token-level loss  
11 interventions fail to address. This observation motivates moving beyond token-  
12 level loss interventions toward trajectory-level output corrections. We propose  
13 Self-Refined Distillation (SRD), a trajectory-level correction method that revises  
14 the student’s rollout under the teacher guidance while within on-policy support.  
15 By correcting problematic prefixes before distillation, SRD mitigates prefix failure  
16 at its source. Moreover, SRD improves exploration by exposing the student to  
17 alternative valid derivations under teacher guidance, even when original rollouts are  
18 already correct. Across a wide range of benchmarks and base models at multiple  
19 scales, SRD consistently outperforms prior baselines, improving single-attempt  
20 accuracy and broadening reasoning coverage.

## 21 1 Introduction

22 On-policy distillation (OPD), which computes per-token teacher supervision along the student’s  
23 own rollouts, has quickly secured a place in modern large language model (LLM) post-training (Gu  
24 et al., 2024; Agarwal et al., 2024; Lu & Thinking Machines Lab, 2025). Recent industry releases  
25 including Qwen3 (Yang et al., 2025), DeepSeek-v4 (DeepSeek-AI, 2026), MiMo-v2 (Xiao et al.,  
26 2026), and GLM-5 (Zeng et al., 2026) all incorporate an OPD stage alongside supervised fine-tuning  
27 (SFT) and reinforcement learning with verifiable rewards (RLVR). While OPD has achieved strong  
28 performance, it typically relies on a distinct teacher model to supervise training. On-policy self-  
29 distillation (OPSD) offers a lightweight alternative: it is a parameter-sharing variant of OPD in which  
30 the teacher and student are the same model under different contexts. The student is conditioned only  
31 on the problem statement, while the teacher is additionally conditioned on privileged information,  
32 such as the ground-truth answer (Zhao et al., 2026; Hübötter et al., 2026; Shenfeld et al., 2026).

33 Despite these successes, recent studies show that vanilla OPD/OPSD recipes often fall short of their  
34 promise, exhibiting failure modes that turn the supervision signal noisy or totally uninformative (Fu  
35 et al., 2026; Xu et al., 2026b). Existing remedies address these issues at the token-loss level,  
36 reweighting or clipping per-token contributions while leaving the sampled trajectory itself unchanged.  
37 For example, Zhao et al. (2026) clip per-token losses above a fixed threshold to suppress destabilizing  
38 high-KL tokens. Unfortunately, by acting only at the per-token loss level, these interventions fail  
39 to mitigate *prefix failure*, an inherent limitation we formalize in this work. Prefix failure happens

when the student’s rollout takes a wrong reasoning path and almost no continuation of that prefix can reach the correct solution without backtracking or reflection. Under prefix failure, the per-token teacher distribution becomes a bimodal mixture between continuing the failed prefix and pivoting back toward the reference (Sec. 4.1); and even granting an ideal teacher, because the per-token KL is evaluated along the student’s *frozen rollout*, the resulting gradient is fragmented and its supervision pairs diverge from the correction path itself (Sec. 4.2). Recovery therefore requires a *trajectory-level* improvement; prior per-token approaches operate at the wrong scale, i.e., token-level, leaving the failed prefix structurally intact.

To mitigate the prefix failure, we propose Self-Refined Distillation (SRD), a simple yet effective trajectory-level refinement strategy that retains on-policy support while incorporating the reference solution as a guidance. We narrow our focus to the OPSD setting, which controls for confounds from teacher–student model mismatch in general OPD. Concretely, for each problem–solution pair, we first sample a raw on-policy rollout and then prompt the teacher model to produce a refined version of that rollout guided by the reference solutions. The refined trajectory is then used as supervision for subsequent OPSD updates with forward KL (Sec. 5). We empirically validate SRD on five competition-math benchmarks (AIME24/25, HMMT25, BeyondAIME, AMOBench) for Qwen3-4B and Qwen3-8B base models. SRD achieves state-of-the-art performance at both scales and delivers strong Pass@16 gains over the base (e.g.,  $\sim 50\%$  relative on AMOBench). Notably, on AMOBench SRD solves 9/23 (39%) of the questions the base model fails on all 16 sampling attempts, nearly doubling the best baseline (22%); these results confirm both single-attempt sharpening and reasoning-coverage expansion.

## 2 Related Work

**On-Policy Distillation.** On-policy distillation (OPD) for post-training LLMs traces back to classical knowledge distillation (Hinton et al., 2015) and replaces the fixed-corpus targets there with per-token teacher supervision computed along the student’s own rollouts. OPD couples with the on-policy sampling property and dense token-level learning signal by token-level KL loss (Gu et al., 2024; Agarwal et al., 2024; Song & Zheng, 2026; Yang et al., 2025; DeepSeek-AI, 2026; Xiao et al., 2026; Zeng et al., 2026). By contrast, RLVR offers only a sparse trajectory-level reward that scales poorly when most rollouts fail the verifier, whereas SFT relies on off-policy reference data and forfeits the on-policy structure that drives compute-efficient learning (Lu & Thinking Machines Lab, 2025).

**On-policy Self-distillation.** On-policy self-distillation (OPSD) is a special case of OPD that instantiates teacher and student from the same model under different privileged contexts, thereby removing the need for a separate teacher and enabling self-improvement without external supervision (Zhao et al., 2026; Hübotter et al., 2026; Shenfeld et al., 2026). The privileged context typically can be involved in reference solutions, feedback, knowledge and experiences, or other auxiliary information that is unavailable to the student (Shi et al., 2026; Ye et al., 2026; Penaloza et al., 2026; Wang et al., 2026; Stein et al., 2026). Operating within the OPD paradigm, OPSD inherits most of the same failure modes, e.g., the privileged supervision can collapse into vacuous guidance as training progresses. A similar concern appears in Zhao et al. (2026), who clip unusually large per-token losses to suppress unreliable learning signals and stabilize training.

**Common Failure Mode and Fix.** Recent studies report that those vanilla distillation methods often underperform in practice and exhibit a range of failure modes, including mode collapse, trajectory inflation, and supervision signals that vanish or even actively mislead the student and more (Fu et al., 2026; Xu et al., 2026b; Luo et al., 2026; Yang et al., 2026a; Kim et al., 2026; Li et al., 2026a; Xu et al., 2026a; Song & Zheng, 2026). Most of these failure modes are addressed through token-level dense-KL loss interventions. For example, Fu et al. (2026) find that the teacher distribution under OPD can be dominated by a small number of high-loss tokens, and propose top- $K$  truncation to restrict supervision to high-confidence tokens. Xu et al. (2026b) upweights informative tokens, e.g., tokens with low student entropy but high teacher–student divergence, to achieve better results.

While the per-token interventions above may look contradictory (clipping high-loss tokens versus amplifying them), they share the goal of selecting informative learning signals, with the specific choice dictated by the divergence choice (Sec. 4.1). These interventions are motivated by empirically observation, yet the empirical failures in OPSD may partly be attributable to *prefix failure* (Sec. 4).

### 3 Preliminaries

**On-policy Distillation.** Knowledge distillation trains a student model  $\pi_S$  to match the output distribution of a teacher  $\pi_T$ , beyond the hard reference label (Hinton et al., 2015). In on-policy distillation (OPD) of LLMs, supervision is computed on trajectories sampled from the current student rather than on fixed expert prefixes (Gu et al., 2024; Agarwal et al., 2024; Lu & Thinking Machines Lab, 2025). Given a prompt  $x \sim \mathcal{D}$  from the training dataset, the student samples an autoregressive rollout  $y \sim \pi_\theta(\cdot | x)$ . The teacher  $\pi_T(\cdot | x, y_{<t})$  is then evaluated on the student-visited prefixes  $y_{<t}$ , producing a dense token-level learning signal. A representative OPD objective minimizes the per-token reverse KL divergence between student and teacher,

$$\mathcal{L}(\theta) = \mathbb{E}_{x \sim \mathcal{D}, y \sim \pi_\theta(\cdot | x)} \left[ \frac{1}{|y|} \sum_{t=1}^{|y|} D(\pi_\theta(\cdot | x, y_{<t}) \| \pi_T(\cdot | x, y_{<t})) \right].$$

**On-policy Self-Distillation.** On-policy self-distillation (OPSD) removes the need for a separate teacher by embodying teacher and student policies from the same model under different contexts (Zhao et al., 2026; Hübottner et al., 2026; Shenfeld et al., 2026). Given a problem-solution pair  $(x, y^*) \sim \mathcal{D}$ , the teacher policy receives privileged information such as the reference answer or reasoning trace and is evaluated as  $\pi_T = \pi_\theta(\cdot | x, y^*, y_{<t})$ , with teacher and student sharing the parameters  $\theta$  of the same model. The standard OPSD loss is

$$\mathcal{L}(\theta) = \mathbb{E}_{(x, y^*) \sim \mathcal{D}, y \sim \pi_\theta(\cdot | x)} \left[ \frac{1}{|y|} \sum_{t=1}^{|y|} D(\pi_\theta(\cdot | x, y_{<t}) \| \text{sg}[\pi_\theta(\cdot | x, y^*, y_{<t})]) \right], \quad (1)$$

where  $\text{sg}[\cdot]$  denotes the stop-gradient operator applied to the teacher branch. The choice of divergence is itself a design. Reverse KL is mode-seeking and is generally preferred for generative language-model distillation since it discourages the student from assigning probability to low-probability regions of the teacher, whereas forward KL is mode-covering (Gu et al., 2024; Lu & Thinking Machines Lab, 2025; Zhao et al., 2026; Jin et al., 2026; Jang et al., 2026). In practice, the full vocabulary  $\mathcal{V}$  is widely adopted to reduce variance and stabilize gradient estimates relative to single-sample Monte Carlo estimation (Zhao et al., 2026; Yang et al., 2025). Having full access to both distributions makes the KL direction independent of the sampling distribution; see Jin et al. (2026) for detailed discussion.

## 4 Prefix Failure in Token-Level On-Policy Self-Distillation

### 4.1 Mixture of Distribution

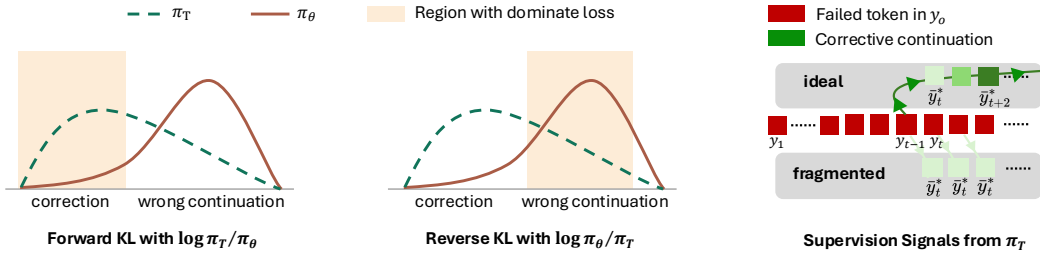


Figure 1: **Left and Middle:** Under prefix failure, the teacher distribution becomes a mixture with two modes. By their respective mode-covering and mode-seeking properties, forward KL is dominated by the correction-onset region, while reverse KL is dominated by the wrong-continuation region. **Right:** Supervision signal provided by the teacher under prefix failure.

Dense per-token KL relies on the teacher providing a reliable supervisory signal at every position along the student rollout. When the teacher faces a wrong reasoning path rolled out by the students, the supervision can turn noisy or even adversarial. Let  $F(y_{o,<t})$  denote the *prefix-failure* event—the prefix  $y_{o,<t}$  contains reasoning errors that contradict  $y^*$  and cannot be extended to  $y^*$  without

retraction or contradiction. Whenever  $F(y_{o,<t})$  holds, the teacher becomes a mixture distribution with two modes, one that continues the failed prefix  $y_{o,<t}$  for sequence consistency and another that pivots back toward  $y^*$ . This mixture structure turns the supposedly dense supervision into a noisy or even adversarial signal. A complementary failure mode arises on degenerate prefixes (e.g., repetition loops), where the teacher instead remains locally aligned with the student and the guidance signal vanishes entirely (Fu et al., 2026, Figure 3). Notably, prefix failure is unique to the on-policy dense signal training paradigm: SFT’s fixed trajectories stay aligned with  $y^*$ , and RLVR’s sparse end-of-trajectory rewards never touch intermediate prefixes.

The choice of KL direction interacts with prefix failure asymmetrically. Recovering from  $F(y_{o,<t})$  requires a corrective continuation  $\bar{y}_{\geq t}^* = (\bar{y}_t^*, \bar{y}_{t+1}^*, \dots, \bar{y}_{t+k-1}^*)^1$ , generated autoregressively along the correction path, where  $\bar{y}_t^*$  is a correction-onset token ( $\bar{y}_t^* \in \{\text{Wait}, \text{Actually}, \dots\}$ ) and  $\bar{y}_{t+1}^*, \dots, \bar{y}_{t+k-1}^*$  continue the recovery, with  $k$  denoting the length of this continuation. Under  $F(y_{o,<t})$ , the ideal teacher partially shifts mass from the natural continuation toward  $\bar{y}_t^*$ , while the student with high probability remains anchored on the wrong continuation. Forward KL  $D(\pi_T \| \pi_\theta)$ , weighted by  $\pi_T$ , is dominated by the correction-onset region; its mode-covering nature forces the student onto this out-of-distribution (OOD) mode. This can destabilize training and, in the worst case, lead to mode collapse (Zhao et al., 2026). In contrast, reverse KL  $D(\pi_\theta \| \pi_T)$  is weighted by  $\pi_\theta$ , so its mode-seeking behavior makes the loss dominated by the wrong-continuation region where the student already places high mass. Since  $\pi_\theta(\bar{y}_t^*)$  is small, the supervision has little effect, causing updates to concentrate on the failing trajectory rather than the recovery tokens.

Prior work has empirically identified various failure modes of dense token-level KL training and proposed loss-design remedies. Many of these failures are consistent with the prefix-failure mechanism analyzed above, even when not labeled as such. Under forward KL, Zhao et al. (2026) clip each per-token loss by a threshold  $c > 0$  to suppress the high-KL tokens that destabilize training. Under reverse KL, Fu et al. (2026) restrict the divergence to the teacher’s top- $K$  vocabulary, and Xu et al. (2026b) reweight per-token losses by entropy and student-teacher disagreement, both aiming to amplify the informative token signal. In effect, those reverse-KL fixes attempt to induce mode-covering behavior within a mode-seeking objective, manually amplifying teacher modes (e.g.,  $\bar{y}_t^*$ ) that mode-seeking would otherwise underweight. These loss-level interventions, however, leave the underlying failed prefix in place. Fig. 1 (Left and Middle) depicts the mixture and KL asymmetry.

## 4.2 Can a Perfect Teacher Recover the Correction Path?

Even granting an ideal teacher, dense per-token KL is structurally limited because it is a per-position objective among student’s rollout. To make this precise, we trace the per-token OPSD loss back to its sequence-level origin, which makes the underlying mechanism more transparent. Define the per-token log-ratio  $\delta_t := \log \pi_\theta(y_t | x, y_{<t}) - \log \pi_T$  with  $\pi_T = \pi_\theta(y_t | x, y^*, y_{<t})$ . Differentiating the sequence-level reverse KL of Eq. (1) (full derivation in Sec. A) yields the policy-gradient form

$$\nabla_\theta \mathcal{J}(\theta) = \mathbb{E}_{\substack{(x, y^*) \sim \mathcal{D} \\ y \sim \pi_\theta(\cdot | x)}} \left[ \sum_{t=1}^{|y|} \left( \delta_t + \sum_{t'=t+1}^{|y|} \delta_{t'} \right) \nabla_\theta \log \pi_\theta(y_t | x, y_{<t}) \right], \quad (2)$$

where  $-\delta_t$  acts as a token-level return. In practice, however, standard OPSD implementations (Lu & Thinking Machines Lab, 2025; Yang et al., 2025) do not optimize the sequence-level KL in Eq. (1); instead, they retain only the immediate log-ratio at each position, yielding the per-token surrogate

$$\nabla_\theta \mathcal{J}(\theta) = \mathbb{E}_{\substack{(x, y^*) \sim \mathcal{D} \\ y \sim \pi_\theta(\cdot | x)}} \left[ \sum_{t=1}^{|y|} \delta_t \nabla_\theta \log \pi_\theta(y_t | x, y_{<t}) \right]. \quad (3)$$

We contrast Eq. (3) with the *perfect teacher* would induce by autoregressively unfolding  $\bar{y}_{\geq t}^*$  along the correction path, delivering the supervision pairs

$$\{(y_{o,<t}, \bar{y}_t^*), ((y_{o,<t}, \bar{y}_t^*), \bar{y}_{t+1}^*), \dots, ((y_{o,<t}, \bar{y}_{t:t+k-1}^*), \bar{y}_{t+k-1}^*)\},$$

in which the context grows along the correction path itself. The corresponding ideal gradient is

$$g_{\text{ideal}} = - \sum_{i=1}^k \delta_i^{\text{ideal}} \cdot \nabla_\theta \log \pi_\theta(\bar{y}_{t+i-1}^* | x, y_{o,<t}, \bar{y}_{t:t+i-1}^*), \quad (4)$$

<sup>1</sup>  $\bar{y}_{\geq t}^*$  is not the literal suffix of  $y^*$ , but any correct continuation conditioned on  $y_{<t}$ , i.e.,  $\text{Verify}(y_{<t} \cdot \bar{y}_{\geq t}^*, y^*) = 1$ .

where  $\delta_i^{\text{ideal}} := \log \pi_\theta(\bar{y}_{t+i-1}^* | x, y_{o,<t}, \bar{y}_{t:t+i-1}^*) - \log \pi_T(\bar{y}_{t+i-1}^* | x, y_{o,<t}, \bar{y}_{t:t+i-1}^*)$ . Yet under dense KL the contexts are dictated by the *frozen* student trajectory, not by the unfolding correction. At position  $t$ , the teacher conveys  $\bar{y}_t^*$  given  $y_{o,<t}$ , and the student updates its parameters to favor  $\bar{y}_t^*$ . At position  $t+1$ , however, the teacher’s supervision is conditioned on  $y_{o,<t+1} = (y_{o,<t}, y_{o,t})$  (the original failed trajectory’s own continuation), not on the correction path  $(y_{o,<t}, \bar{y}_t^*)$ . Because every subsequent prefix the teacher sees is still anchored in the original failure rather than the unfolding correction, the teacher is left repeatedly recommending the same correction-onset token. The supervision pairs delivered to the student therefore form the fragmented sequence

$$\{(y_{o,<t}, \bar{y}_t^*), (y_{o,<t+1}, \bar{y}_t^*), \dots, (y_{o,<t+k-1}, \bar{y}_t^*)\},$$

in which the context grows along the wrong continuation while the target remains stuck at  $\bar{y}_t^*$ . The corresponding gradient

$$g_{\text{frag}} = - \sum_{i=0}^{k-1} \delta_i^{\text{frag}} \cdot \nabla_\theta \log \pi_\theta(\bar{y}_t^* | x, y_{o,<t+i}),$$

with  $\delta_i^{\text{frag}} := \log \pi_\theta(\bar{y}_t^* | x, y_{o,<t+i}) - \log \pi_T(\bar{y}_t^* | x, y_{o,<t+i})$ , evaluates the score function at a completely different set of (context, token) pairs than  $g_{\text{ideal}}$ . The two pair sets share only their first element  $(y_{o,<t}, \bar{y}_t^*)$ ; beyond it,  $g_{\text{ideal}}$  propagates supervision along  $(y_{o,<t}, \bar{y}_t^*, \bar{y}_{t+1}^*, \dots)$  while  $g_{\text{frag}}$  accumulates supervision along  $(y_{o,<t}, y_{o,t}, y_{o,t+1}, \dots)$ . The two trajectories diverge after a single step and never re-intersect:

$$\underbrace{\{((y_{o,<t}, \bar{y}_{t:t+i-1}^*), \bar{y}_{t+i-1}^*)\}_{i=1}^k}_{\text{required by recovery}} \cap \underbrace{\{(y_{o,<t+i}, \bar{y}_t^*)\}_{i=0}^{k-1}}_{\text{delivered}} = \{(y_{o,<t}, \bar{y}_t^*)\}.$$

The privileged information  $y^*$  thus stays trapped in per-position marginals. Dense KL keeps recommending  $\bar{y}_t^*$  on ever-deepening wrong-continuation contexts but cannot supervise the multi-step unfolding of  $\bar{y}_{\geq t}^*$ , and loss-level interventions only reweight terms within the visited pair set  $\{(y_{o,<t+i}, \bar{y}_t^*)\}$  rather than move the gradient onto the correction-path pair set. The  $g_{\text{frag}}$  signal is not useless, however; biasing the student toward  $\bar{y}_t^*$  can trigger self-correction, though unfolding the full  $\bar{y}_{\geq t}^*$  is bounded by the student’s capacity. Our method SRD (Sec. 5) recovers  $g_{\text{ideal}}$  by supervising the per-token KL along a refined trajectory generated under privileged context, so the supervision contexts grow along the correction path itself rather than the frozen failed prefix.

### 4.3 Experimental Validation of Prefix Failure

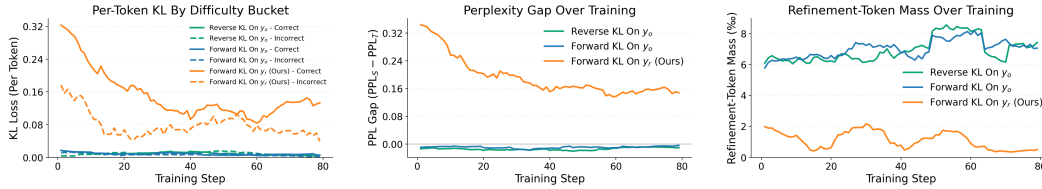


Figure 2: Empirical observations of prefix failure under standard OPSD ( $y_o$ , forward and reverse KL). **Left:** Per-token KL by correct/incorrect rollouts. **Middle:** Teacher-student perplexity gap. **Right:** Teacher’s epistemic mass.

We empirically verify the prefix failure mechanism through three measurements over OPSD training on student rollouts  $y_o$  under both forward and reverse KL (Fig. 2). A third curve (*ours*) is included for reference, corresponding to the method introduced in Sec. 5.

**Distribution-level supervision collapse** (left). We split the per-token KL between  $\pi_T$  and  $\pi_S$  by stage-1 verifier outcome (correct vs. incorrect) into token-weighted means  $D^{\text{correct}}$  and  $D^{\text{incorrect}}$ . Both stay pinned near zero on  $y_o$ , so  $\pi_T$  and  $\pi_S$  remain aligned and dense KL delivers no signal once  $F(y_{o,<t})$  saturates. Notably, even  $D^{\text{incorrect}}$  stays near zero, indicating that the teacher with high probability collapses onto the student’s failure rather than diverging to correct it.

**Perplexity gap collapse** (middle). We measure teacher and student perplexities token-wise on the same response mask under teacher-forced decoding,  $\text{PPL}_S = \exp(-1/|y_o| \sum_t \log \pi_S(y_{o,t} | y_{o,<t}))$



and  $\text{PPL}_T$  defined analogously with  $\pi_T$ . The gap  $\text{PPL}_S - \text{PPL}_T$  stays near zero throughout training, so the privileged condition  $y^*$  delivers essentially vanishing incremental supervision over what the student already represents on its own rollouts.

**Epistemic-token mass** (right). The teacher places 6–8% of its per-position mass on 16 epistemic onset tokens throughout training, matching the  $\bar{y}_t^*$ -repeat signature of  $g_{\text{frag}}$  predicted in Sec. 4.2. Strikingly, student and teacher top-16 tokens already absorb 97 – 99 % of total probability mass (Li et al., 2026b, Figure 18), so this allocation commands a disproportionately dominant share of the remaining 1 – 3 % residual budget. The same metric on  $y_r$  (*ours*) collapses below 2%, confirming concentration is tied to failed-prefix conditioning rather than a universal teacher property.

## 5 Self-Refined Distillation

The previous section identifies *prefix failure* as one of the central bottlenecks in OPSD. With prefix failure, dense supervision becomes noisy and can even fail to guide the student. Most existing mitigations operate through loss design and leave the offending prefix unchanged. Directly optimizing prefix failure is intractable since identifying the failure index requires enumerating  $\mathcal{O}(|\mathcal{V}|^k)$  with following reasoning paths of length  $k$ . We therefore relax the target to a trajectory-level surrogate that maximizes the expected verifier-pass rate over the dataset  $\mathcal{D}$  under the support constraint of  $\pi_\theta$ :

$$\max_q \mathbb{E}_{(x, y^*) \sim \mathcal{D}} [\Pr_{y \sim q(\cdot | x, \cdot)} \{\text{Verify}(y, y^*) = 1\}] \quad \text{s.t.} \quad \pi_\theta(y | x) > 0 \quad (5)$$

We note that this is a *trajectory-level* distribution support constraint and the optimization objective form mirrors a standard RLVR objective. We emphasize, however, that  $\theta$  is *not* the direct optimization variable: the equation instead specifies the goal of an upstream process before the OPSD gradient update with  $\theta$ , namely to construct trajectories that attain higher verifier-pass rates while remaining within the current student’s support. These trajectories then serve as the supervision for the subsequent update of the OPSD through the loss of the standard distribution matching in Eq. (1).

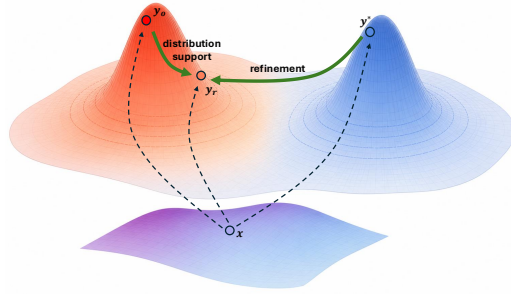
Two extreme choices of  $q$  illustrate the tension between objective and constraint. (I) Setting  $q(\cdot | x, \cdot) = \pi_\theta(\cdot | x)$  fails back to standard OPSD. The on-policy constraint is satisfied by construction, but this choice fails to mitigate *prefix failure* beyond current OPSD algorithms, since the supervision trajectories are still drawn from  $\pi_\theta$ . Repeated sampling refines this choice by drawing rollouts and retaining only the verifier-passing ones (Brown et al., 2024; Stein et al., 2026), straining inference budgets linearly and yielding nothing on questions the student cannot already solve. Any improvement in pass rates depends entirely on the stochasticity of base LLM decoding, with no direct optimization of Eq. (5). (II) Setting  $q(\cdot | x, \cdot) = \pi^*$ , i.e., the expert policy that produces  $y^*$ , attains  $\text{Verify} = 1$  exactly but generally violates the on-policy support constraint, so this choice falls outside the feasible set and breaks the on-policy character that OPSD depends on.

To move beyond the two extremes above, we propose *Self-Refined Distillation* (SRD), which operates at the trajectory level to optimize Eq. (5) by first drawing a raw on-policy rollout  $y_o \sim \pi_\theta(\cdot | x)$ , then constructing a refined trajectory  $y_r$  via:

$$y_r \sim q(\cdot | x, \cdot) := \pi_\theta(\cdot | x, y_o, y^*).$$

Crucially, conditioning on  $y_o$  anchors  $y_r$  to the reasoning patterns  $\pi_\theta$  has already demonstrated, i.e., within the policy support, while  $\pi_\theta$  guided by  $y^*$  rewrites the erroneous portions to directly mitigate prefix failure. To our knowledge, this is the first design that explicitly optimizes Eq. (5) while respecting the on-policy constraint without prohibitive additional compute overhead. A refined trajectory  $y_r \sim q(\cdot | x, y_o, y^*)$  is then used as supervision for the subsequent OPSD update. Per-token KL along  $y_r$  reduces exposure to the bimodal teacher mixture (Sec. 4.1) and recovers the ideal gradient  $g_{\text{ideal}}$  (Sec. 4.2), since supervision contexts grow along the refined trajectory rather than the raw rollout  $y_o$  at higher risk of prefix failure.

SRD also boosts the student’s exploration beyond standard OPSD. On a correct  $y_o$ , standard OPSD provides little new signal: it tends to merely reinforces the high-probability solution path the student already produces due to the fragmented gradient. In contrast,  $y_r$  is drawn under the privileged condition  $y^*$  and surfaces alternative valid derivations of the same answer, i.e., paths consistent with  $y^*$  but rarely sampled from  $\pi_\theta(\cdot | x)$  alone, thereby expanding the set of correct reasoning trajectories the student is supervised on. SRD therefore adds value in both regimes: mitigating failed prefixes when  $y_o$  is wrong, and broadening the supervision distribution when  $y_o$  already succeeds.



#### Algorithm 1: Self-Refined Distillation

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**Require:** minibatch  $\mathcal{B} \subset \mathcal{D}$ , policy  $\pi_\theta$

- 1: **for**  $(x, y^*) \in \mathcal{B}$  **do**
- 2:    $y_o \sim \pi_\theta(\cdot | x)$
- 3:    $y_r \sim \pi_\theta(\cdot | x, y_o, y^*)$
- 4:   Update  $\theta$  by Eq. (6) on  $y_r$
- 5: **end for**

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Figure 3: **Left:** SRD refines  $y_o$  to  $y_r$  under privileged context  $y^*$ . **Right:** SRD pseudocode.

249 The training-data analysis in Sec. 6.4 confirms this (e.g., the correct subset of  $y_r$  exhibits a low-length  
 250 mode absent in  $y_o$ ) and translates into Pass@ $k$  gains in Sec. 6.3.

251 Concretely, given  $y_r$ , we instantiate the distillation loss as the forward KL with full vocabulary  
 252 matching over  $\mathcal{V}$ , which provides mode-covering supervision and stabilizes gradient estimates:

$$\mathcal{L}(\theta) = \mathbb{E}_{\substack{(x, y^*) \sim \mathcal{D} \\ y_o \sim \pi_\theta(\cdot | x) \\ y_r \sim \pi_\theta(\cdot | x, y_o, y^*)}} \left[ \frac{1}{|y_r|} \sum_{t=1}^{|y_r|} D(\text{sg}[\pi_\theta(\cdot | x, y_o, y^*, y_{r, < t})] \parallel \pi_\theta(\cdot | x, y_{r, < t})) \right], \quad (6)$$

253 The exact prompt template is given in Sec. C.4; Fig. 3 and Algorithm 1 illustrate the full procedure.  
 254 Fig. 2 confirms that these design choices alleviate the failures observed on  $y_o$ . Beyond the per-token  
 255 KL recovery on  $y_r$  (Left, discussed above), the teacher-student perplexity gap opens on  $y_r$  (Middle),  
 256 restoring the incremental supervision, and the teacher’s epistemic onset mass decreases 3x less than  
 257  $y_o$ . KL and the PPL gap both decrease steadily as epistemic onset concentration fades, indicating that  
 258 the privileged signal is genuinely transmitted to the student. The flat  $y_o$  curves, in contrast, reflect  
 259 collapsed learning signals under prefix failure.

## 6 Experiments

260 We evaluate SRD against four dense-KL baselines on five competition-math benchmarks. We answer  
 261 three questions. (I) Does SRD improve exploitation (Avg@16) over the base model (Sec. 6.2)?  
 262 (II) Does SRD improve exploration (Pass@16) beyond the base model’s support (Sec. 6.3)? (III)  
 263 How does  $y_r$  differ from  $y_o$  on the training corpus (Sec. 6.4)? Sec. B.1 provides a detailed training-  
 264 trajectory analysis comparing  $y_o$  and  $y_r$  across model scales and reference-quality regimes; Sec. B.2  
 265 stratifies AMOBench test rollouts by base-model difficulty bucket to localize where SRD’s Pass@16  
 266 gain originates; Sec. B.3 ablates the training corpus by initial-rollout correctness on Qwen3-8B. Full  
 267 experimental details are given in Sec. C.

### 6.1 Experiments Setup and Baselines

270 **Models and training data.** We use Qwen3-4B-Instruct-2507 and Qwen3-8B (both with thinking  
 271 enabled) as base models (Yang et al., 2025), and train on the DeepScaleR corpus of roughly 40K  
 272 math problems with verified answers (Luo et al., 2025). Following the OPSD setup (Sec. 3), the  
 273 same backbone embodies teacher and student via the privileged context  $y^*$ , with no separate teacher  
 274 network. For every prompt  $x$ , we first draw a raw rollout  $y_o \sim \pi_\theta(\cdot | x)$  from the current student;  
 275 for SRD we additionally draw the refined trajectory  $y_r$  via the rewrite prompt in Sec. C.4, while the  
 276 dense-KL baselines train directly on  $y_o$ . We use LoRA adapters (rank 64,  $\alpha=128$ ) on all attention  
 277 and MLP linear layers and train for one epoch with bfloat16 and gradient checkpointing. Full  
 278 hyperparameters and the data-generation pipeline are given in Sec. C.

279 **Baselines.** All baselines minimize per-token KL between the student and the privileged-context  
 280 teacher over the full Qwen3 vocabulary ( $|\mathcal{V}| \approx 152\text{K}$ ), differing only in KL direction and the  
 281 per-token correction applied to mitigate prefix-failure symptoms. *Forward KL* ( $D(\pi_T \parallel \pi_\theta)$ ) is

Table 1: Avg@16 results (%). Per-question average over 16 sampled completions. Parentheses report absolute percentage-point changes from the corresponding base model. Bold marks best performance within each model block, and red cells mark drops below the corresponding base model.

| Method                                      | Traj. | AIME24             | AIME25             | HMMT25             | BeyondAIME         | AMOBench           |
|---------------------------------------------|-------|--------------------|--------------------|--------------------|--------------------|--------------------|
| <i>Qwen3-4B-Instruct-2507 (w/ thinking)</i> |       |                    |                    |                    |                    |                    |
| Base                                        | —     | <b>63.1</b>        | 48.2               | 31.3               | 32.2               | 10.6               |
| +Forward KL                                 | $y_o$ | 60.3 (-2.8)        | 48.8 (+0.6)        | 30.8 (-0.5)        | 32.3 (+0.1)        | 9.5 (-1.1)         |
| +Forward KL w/ Clip                         | $y_o$ | 62.1 (-1.0)        | 49.2 (+1.0)        | 27.9 (-3.4)        | 32.4 (+0.2)        | 9.3 (-1.3)         |
| +Reverse KL                                 | $y_o$ | 58.4 (-4.7)        | 48.8 (+0.6)        | 30.0 (-1.3)        | 31.6 (-0.6)        | 10.1 (-0.5)        |
| +Reverse KL w/ Top-K                        | $y_o$ | 63.0 (-0.1)        | 49.0 (+0.8)        | 31.0 (-0.3)        | 32.3 (+0.1)        | 9.6 (-1.0)         |
| +SRD (ours)                                 | $y_r$ | <b>63.1</b> (+0.0) | <b>49.4</b> (+1.2) | <b>32.1</b> (+0.8) | <b>32.7</b> (+0.5) | <b>11.2</b> (+0.6) |
| <i>Qwen3-8B (w/ thinking)</i>               |       |                    |                    |                    |                    |                    |
| Base                                        | —     | <b>76.5</b>        | 66.7               | 41.5               | 41.6               | 15.9               |
| +Forward KL                                 | $y_o$ | 74.8 (-1.7)        | 65.4 (-1.3)        | 40.3 (-1.2)        | 39.6 (-2.0)        | 15.7 (-0.2)        |
| +Forward KL w/ Clip                         | $y_o$ | <b>76.5</b> (+0.0) | 68.3 (+1.6)        | 42.5 (+1.0)        | 40.2 (-1.4)        | 15.9 (+0.0)        |
| +Reverse KL                                 | $y_o$ | 75.4 (-1.1)        | 68.2 (+1.5)        | 44.3 (+2.8)        | 40.8 (-0.8)        | 16.8 (+0.9)        |
| +Reverse KL w/ Top-K                        | $y_o$ | 75.6 (-0.9)        | 68.5 (+1.8)        | 44.4 (+2.9)        | 41.9 (+0.3)        | 15.2 (-0.7)        |
| +SRD (ours)                                 | $y_r$ | <b>76.5</b> (+0.0) | <b>69.2</b> (+2.5) | <b>44.5</b> (+3.0) | <b>42.8</b> (+1.2) | <b>17.3</b> (+1.4) |

the unmodified mode-covering baseline. *Forward KL w/ Clip* caps each token’s KL contribution at  $c=0.06$ , reproducing the OPSD recipe of Zhao et al. (2026). *Reverse KL* ( $D(\pi_\theta \parallel \pi_T)$ ) is the unmodified mode-seeking baseline. *Reverse KL w/ Top-K* restricts the loss to the teacher’s top-32 support and renormalizes both distributions inside it, reproducing the local-support-matching objective of Fu et al. (2026). All four baselines train on the raw on-policy rollout  $y_o \sim \pi_\theta(\cdot \mid x)$ ; SRD instead trains on the refined trajectory  $y_r$  defined in Sec. 5.

**Evaluation.** We report Avg@16 and Pass@16 on AIME24 (Mathematical Association of America, 2024), AIME25 (Mathematical Association of America, 2025), HMMT25 (HMMT, 2025), BeyondAIME (Seed, 2025), and AMOBench (An et al., 2025), following the Qwen3 evaluation setup (Yang et al., 2025). For each test question we draw  $K=16$  completions and grade them with an external verifier; Avg@16 averages the  $K$  binary outcomes per question, and Pass@16 is the per-question indicator that at least one of the  $K$  samples is correct, both then averaged over the test set. Formal definitions are in Sec. C.2.

## 6.2 SRD Sharpens Single-Attempt Accuracy

Tab. 1 reports Avg@16. SRD is best on every benchmark at both scales and never drops below base (no red cells in SRD rows). AIME24 and AIME25 are largely saturated at this scale (SRD matches base on AIME24, all methods within  $\sim 2$ pp on AIME25), so the contrast with loss-design baselines is sharpest on the other three benchmarks. Three of four baselines regress on at least one benchmark (e.g., Reverse KL  $-5.0$ pp on AIME24-4B, Forward KL w/ Clip  $-3.4$ pp on HMMT25-4B, Forward KL  $-2.0$ pp on BeyondAIME-8B), reflecting the prefix-failure asymmetry of Sec. 4, while SRD delivers consistent gains (e.g.,  $+3.0$ pp on HMMT25 and  $+1.4$ pp on AMOBench at 8B). Notably, the regressions span both KL directions and both per-token corrections (clipping and top- $K$  truncation), indicating that prefix-failure-induced drops cannot be repaired by tuning the loss alone. On Qwen3-4B in particular, every dense-KL baseline regresses on at least one of HMMT25/BeyondAIME/AMOBench, whereas SRD delivers  $+0.5$  to  $+1.2$ pp gains on every non-saturated column. Fixing the offending trajectory is more reliable than reweighting the loss it induces.

## 6.3 SRD Expands Reasoning Coverage

Tab. 2 reports Pass@16, where SRD’s trajectory-level rewrite separates most clearly from per-token interventions. On Qwen3-8B, SRD lifts AMOBench from 41.0% to 61.5% ( $+20.5$ pp, a 50% relative gain) and HMMT25 from 66.7% to 73.3% ( $+6.6$ pp), while the strongest dense-KL baseline on AMOBench stops at 51.3% and three of four baselines drop on AIME24. On Qwen3-4B-Instruct-2507, SRD posts  $+5.0$ pp on BeyondAIME and  $+10.7$ pp on AMOBench, again top of every column. On AMOBench, SRD solves 9/23 (39%) of the questions the base fails on all 16 attempts, nearly doubling the best baseline (+Forward KL at 22%); these are unreachable for the base at  $K=16$ , so the gain is genuine support expansion. See more detailed test rollout analysis in Sec. B.2.



Table 2: Pass@16 results (%). Parentheses report absolute percentage-point changes from the corresponding base model. Bold marks best performance within each model block, and red cells mark drops below the corresponding base model.

| Method                                      | Traj. | AIME24             | AIME25             | HMMT25             | BeyondAIME         | AMOBench            |
|---------------------------------------------|-------|--------------------|--------------------|--------------------|--------------------|---------------------|
| <i>Qwen3-4B-Instruct-2507 (w/ thinking)</i> |       |                    |                    |                    |                    |                     |
| Base                                        | –     | 83.3               | 76.7               | 50.0               | 59.0               | 23.1                |
| +Forward KL                                 | $y_o$ | <b>86.7</b> (+3.4) | 76.7 (+0.0)        | 53.3 (+3.3)        | 58.0 (-1.0)        | 33.3 (+10.2)        |
| +Forward KL w/ Clip                         | $y_o$ | <b>86.7</b> (+3.4) | 76.7 (+0.0)        | 46.7 (-3.3)        | 62.0 (+3.0)        | 28.2 (+5.1)         |
| +Reverse KL                                 | $y_o$ | <b>86.7</b> (+3.4) | 76.7 (+0.0)        | 50.0 (+0.0)        | 60.0 (+1.0)        | 28.2 (+5.1)         |
| +Reverse KL w/ Top-K                        | $y_o$ | <b>86.7</b> (+3.4) | 76.7 (+0.0)        | <b>56.7</b> (+6.7) | 57.0 (-2.0)        | 25.6 (+2.5)         |
| +SRD (ours)                                 | $y_r$ | <b>86.7</b> (+3.4) | <b>80.0</b> (+3.3) | <b>56.7</b> (+6.7) | <b>64.0</b> (+5.0) | <b>33.8</b> (+10.7) |
| <i>Qwen3-8B (w/ thinking)</i>               |       |                    |                    |                    |                    |                     |
| Base                                        | –     | <b>90.0</b>        | 83.3               | 66.7               | 66.0               | 41.0                |
| +Forward KL                                 | $y_o$ | 86.7 (-3.3)        | 82.7 (-0.6)        | <b>73.3</b> (+6.6) | 68.0 (+2.0)        | 51.3 (+10.3)        |
| +Forward KL w/ Clip                         | $y_o$ | <b>90.0</b> (+0.0) | <b>86.7</b> (+3.4) | 70.0 (+3.3)        | 61.0 (-5.0)        | 43.6 (+2.6)         |
| +Reverse KL                                 | $y_o$ | 86.7 (-3.3)        | <b>86.7</b> (+3.4) | <b>73.3</b> (+6.6) | 64.0 (-2.0)        | 51.3 (+10.3)        |
| +Reverse KL w/ Top-K                        | $y_o$ | 86.7 (-3.3)        | 86.6 (+3.3)        | 66.6 (-0.1)        | <b>69.0</b> (+3.0) | 38.5 (-2.5)         |
| +SRD (ours)                                 | $y_r$ | <b>90.0</b> (+0.0) | <b>86.7</b> (+3.4) | <b>76.3</b> (+6.6) | 68.0 (+2.0)        | <b>61.5</b> (+20.5) |

## 6.4 Training Trajectory Analysis

Fig. 4 contrasts  $y^*$ , the raw rollout  $y_o$ , and the refined trajectory  $y_r$  on Qwen3-8B over the training corpus, optimizing Eq. (5) indirectly within support. The verifier-pass rate improves from 65.8% on  $y_o$  to 81.4% on  $y_r$  (+15.6pp), and the length distribution compresses by roughly  $9\times$  (median 7.7K  $\rightarrow$  0.88K) toward the reference scale ( $y^*$  median  $\sim$  0.49K), with  $y_r$  remaining within the student’s support since it is sampled from  $\pi_\theta$  itself. We highlight that the same  $\sim 9\times$  compression also applies to the *correct* half of  $y_o$ , surfacing a low-length mode in  $y_r$  that  $y_o$  does not produce. This means even on questions the student already solves, SRD exposes it to alternative, shorter derivations under  $y^*$  guidance, an additional source of supervision diversity; see Sec. B.1 for the full analysis. The same compression also cuts training wall-clock by roughly 60% on Qwen3-8B, which offsets the extra  $y_r$  sampling cost (Sec. C.3). By supervising on  $y_r$  rather than the raw rollout  $y_o$ , SRD also softens the length-inflation regime that Luo et al. (2026) document for OPD training on long on-policy rollouts.

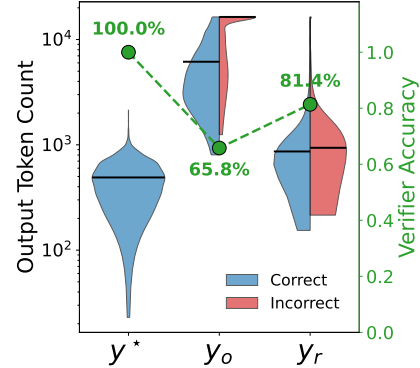


Figure 4: Length distribution and verifier accuracy on Qwen3-8B.

## 7 Conclusion

We identify *prefix failure* as a structural limitation of on-policy (self)-distillation paradigms, where the per-token KL evaluated along the student’s frozen rollout induces a bimodal teacher mixture and a fragmented gradient that loss-level fixes leave structurally intact. To address it, we propose SRD, a trajectory-level refinement that draws a refined trajectory  $y_r$  under privileged context and supervises the per-token KL along  $y_r$ , recovering the ideal supervision-pair structure while remaining on-policy. Across five competition-math benchmarks for Qwen3-4B and Qwen3-8B, SRD attains the best Avg@16 on every benchmark and substantial Pass@16 gains, including solving 9/23 of base-unreachable AMOBench questions and nearly doubling the strongest baseline.

**Limitations.** First, SRD adds one extra sampling pass per problem to construct  $y_r$ . This overhead is largely offset by the roughly  $9\times$  length compression of  $y_r$  relative to  $y_o$ , which proportionally reduces the per-step training compute; on Qwen3-8B the end-to-end wall-clock matches the dense-KL baselines (Sec. C.3). Second, SRD relies on the base model having sufficient capacity to produce a correct continuation under privileged context and thus improve Eq. (5). When  $\pi_\theta(\cdot | x, y_o, y^*)$  cannot reach a verifier-passing trajectory,  $y_r$  collapses toward  $y_o$  and the refinement signal vanishes, leaving SRD with no advantage over baselines in that regime.

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## Appendix: Self-Refined Distillation

### A Derivation of the OPD Policy Gradient

We derive the policy-gradient form [Eq. \(3\)](#) of the on-policy distillation gradient, expressing it in the dense form used in our analysis. The derivation parallels [Yang et al. \(2026b\)](#) and is reproduced here for completeness using the notation of [Sec. 3](#). Throughout,  $\pi_T$  denotes the teacher; under OPSD,  $\delta_t := \log \pi_\theta(y_t | x, y_{<t}) - \log \pi_T$  with  $\pi_T = \pi_\theta(y_t | x, y^*, y_{<t})$ .

**Step 1 (KL as a log-ratio expectation).** Starting from the sequence-level reverse KL  $\mathcal{J}(\theta) = \mathbb{E}[D(\pi_\theta(y | x) || \pi_T(y | x))]$ , the objective expands to

$$\mathcal{J}(\theta) = \mathbb{E}_{x \sim \mathcal{D}, y \sim \pi_\theta(\cdot | x)} [\log \pi_\theta(y | x) - \log \pi_T(y | x)],$$

where the dependence on  $\theta$  enters through both the sampling distribution and the integrand.

**Step 2 (product rule and score-function trick).** Differentiating under the expectation gives

$$\begin{aligned} \nabla_\theta \mathcal{J}(\theta) &= \mathbb{E}_x \left[ \sum_y (\nabla_\theta \pi_\theta(y | x)) (\log \pi_\theta(y | x) - \log \pi_T(y | x)) \right. \\ &\quad \left. + \sum_y \pi_\theta(y | x) \nabla_\theta \log \pi_\theta(y | x) \right]. \end{aligned}$$

The second sum vanishes because

$$\sum_y \pi_\theta(y | x) \nabla_\theta \log \pi_\theta(y | x) = \sum_y \nabla_\theta \pi_\theta(y | x) = \nabla_\theta 1 = 0.$$

Using  $\nabla_\theta \pi_\theta = \pi_\theta \nabla_\theta \log \pi_\theta$ , the remaining term becomes

$$\nabla_\theta \mathcal{J}(\theta) = \mathbb{E}_{x \sim \mathcal{D}, y \sim \pi_\theta(\cdot | x)} [(\log \pi_\theta(y | x) - \log \pi_T(y | x)) \nabla_\theta \log \pi_\theta(y | x)]. \quad (7)$$

**Step 3 (autoregressive decomposition).** Factor  $\log \pi_\theta(y | x) = \sum_t \log \pi_\theta(y_t | x, y_{<t})$  and similarly for  $\pi_T$ . [Eq. \(7\)](#) expands to

$$\mathbb{E}_{x,y} \left[ \sum_{t=1}^{|y|} \sum_{t'=1}^{|y|} \underbrace{(\log \pi_\theta(y_{t'} | x, y_{<t'}) - \log \pi_T(y_{t'} | x, y_{<t'}))}_{\delta_{t'}} \nabla_\theta \log \pi_\theta(y_t | x, y_{<t}) \right].$$

**Step 4 (causality, future tokens do not contribute).** For any  $t' < t$ ,  $\delta_{t'}$  is measurable with respect to  $(x, y_{<t})$ , and conditioning on this prefix yields

$$\mathbb{E}_{y_t \sim \pi_\theta(\cdot | x, y_{<t})} [\nabla_\theta \log \pi_\theta(y_t | x, y_{<t})] = \sum_{y_t} \nabla_\theta \pi_\theta(y_t | x, y_{<t}) = \nabla_\theta 1 = 0,$$

so all cross terms with  $t' < t$  vanish in expectation.

**Step 5 (final form).** Retaining only  $t' \geq t$  recovers [Eq. \(3\)](#),

$$\nabla_\theta \mathcal{J}(\theta) = \mathbb{E}_{x,y} \left[ \sum_{t=1}^{|y|} \left( \sum_{t'=t}^{|y|} \delta_{t'} \right) \nabla_\theta \log \pi_\theta(y_t | x, y_{<t}) \right].$$

The bracketed quantity  $-\sum_{t' \geq t} \delta_{t'}$  acts as a return-to-go for token  $y_t$ . Following common practice ([Agarwal et al., 2024](#); [Lu & Thinking Machines Lab, 2025](#)), applying a discount factor of 0 retains only the term at  $t' = t$ , which gives the per-token surrogate

$$\nabla_\theta \mathcal{J}(\theta) \approx \mathbb{E}_{x,y} \left[ \sum_{t=1}^{|y|} (\log \pi_\theta(y_t | x, y_{<t}) - \log \pi_T(y_t | x, y_{<t})) \nabla_\theta \log \pi_\theta(y_t | x, y_{<t}) \right],$$

which is the gradient of the per-token KL loss in [Sec. 3](#), and, with the privileged-context substitution  $\pi_T(\cdot | x, y_{<t}) = \pi_\theta(\cdot | x, y^*, y_{<t})$ , of the OPSD loss in [Eq. \(1\)](#).

## B Additional Experiments

This section provides three complementary analyses beyond the dataset-averaged numbers in Tabs. 1 and 2. Sec. B.1 characterizes the training corpus by comparing  $y_o$  and  $y_r$  along length, verifier accuracy, and joint outcome on DeepScaleR (Qwen3-4B and Qwen3-8B, with-CoT and without-CoT subsets). Sec. B.2 drills into AMOBench at test time, decomposing Avg@16 and Pass@16 by base-difficulty bucket to localize where SRD’s gains arise. Sec. B.3 ablates Forward-KL, Reverse-KL, and SRD under *fail*, *succ*, and *fail*→*succ* corpus filters on Qwen3-8B.

### B.1 Training-Trajectory Analysis: $y_o$ vs. $y_r$

Here we inspect the training data on which SRD itself is trained, i.e., the raw rollout  $y_o \sim \pi_\theta(\cdot | x)$  and the refined trajectory  $y_r \sim \pi_\theta(\cdot | x, y_o, y^*)$  on DeepScaleR. The reference  $y^*$  supplied with each problem comes in two qualities: a small subset ( $n=4,419$ ) carries a usable reference chain-of-thought, while the rest ( $n=35,826$ ) carries only a short answer-style reference. We split the analysis along this axis to show that SRD’s behavior is consistent across both regimes, reporting Qwen3-4B-Instruct-2507 in Fig. 5 and Qwen3-8B in Fig. 6, with the with-CoT subset on the top row of each figure and the without-CoT subset on the bottom row. We report numbers as 4B / 8B and as with-CoT / without-CoT when the two subsets diverge.

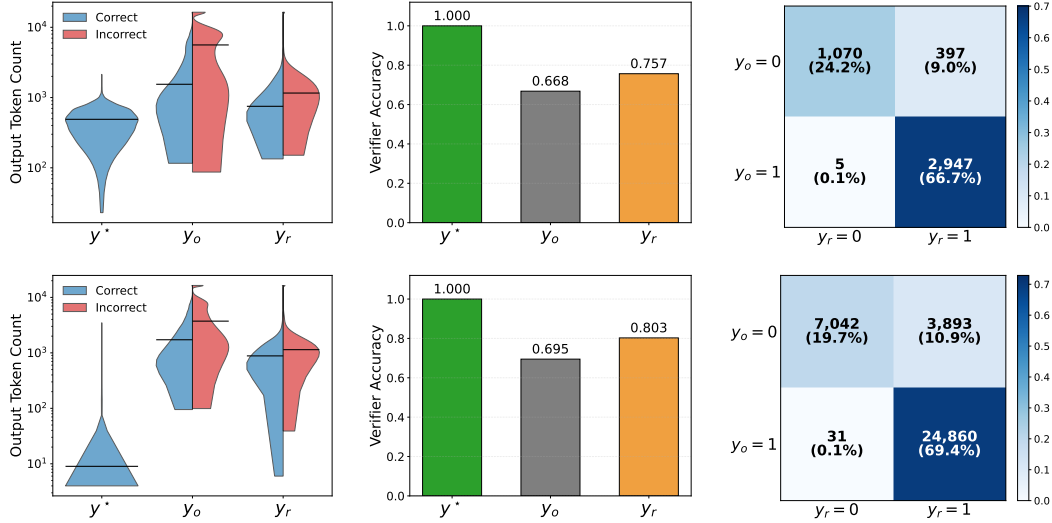


Figure 5: Training-trajectory analysis on Qwen3-4B-Instruct-2507. **Top row:** with-CoT subset ( $n=4,419$ ). **Bottom row:** Without-CoT subset ( $n=35,826$ ). **Left:** Length distribution of  $y^*$ ,  $y_o$ ,  $y_r$ . With  $y_o$  and  $y_r$  split into correct (left) and incorrect (right) halves. **Middle:** Verifier accuracy of the three trajectories. **Right:** Joint outcome of  $y_o$  and  $y_r$  ( $2 \times 2$  confusion); fail→pass cells reflect prefix-failure recovery, pass→fail cells quantify the price.

**Refinement compresses the trajectory.** The left panels show that  $y_r$  moves substantially below  $y_o$  in both regimes. With CoT,  $y^*$  has median  $\sim 0.49$ K tokens and  $y_r$  collapses to 0.85K / 0.88K from  $y_o$ ’s 2.2K / 7.7K (4B / 8B); without CoT,  $y^*$  shrinks to  $\sim 9$  tokens (answer-only) and  $y_r$  lands at 0.93K / 0.83K from  $y_o$ ’s 2.1K / 7.5K. The compression factor on 8B is similar in both regimes ( $\sim 9\times$ ), confirming that conditioning on  $y^*$  pulls the student toward more concise derivations even when the reference is just a short answer string.

**Refinement raises verifier accuracy.** The middle panels report verifier accuracy. With CoT,  $y_o$  passes on 66.8% / 65.8% while  $y_r$  passes on 75.7% / 81.4% (+8.9pp / +15.6pp). Without CoT,  $y_o$  passes on 69.5% / 69.8% and  $y_r$  on 80.3% / 79.8% (+10.8pp / +10.0pp). SRD’s training data therefore contains a substantially higher fraction of correct trajectories than what dense-KL baselines train on across both regimes.



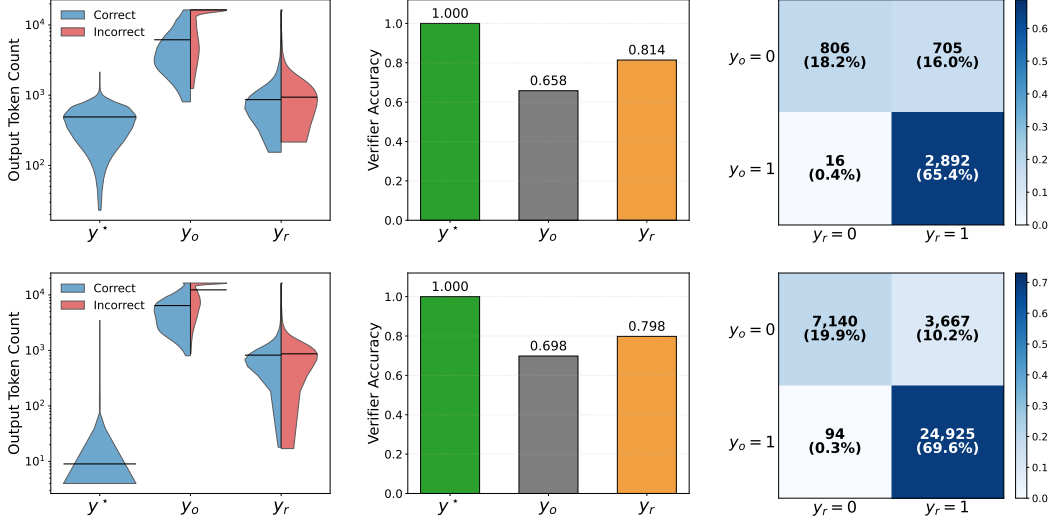


Figure 6: Training-trajectory analysis on Qwen3-8B. **Top row:** with-CoT subset ( $n=4,419$ ). **Bottom row:** Without-CoT subset ( $n=35,826$ ). **Left:** Length distribution of  $y^*$ ,  $y_o$ ,  $y_r$ , with  $y_o$  and  $y_r$  split into correct (left) and incorrect (right) halves. **Middle:** Verifier accuracy of the three trajectories. **Right:** Joint outcome of  $y_o$  and  $y_r$  ( $2 \times 2$  confusion); fail→pass cells reflect prefix-failure recovery, pass→fail cells quantify the price.

**Refinement is monotonically corrective.** The right panels decompose the accuracy gap by joint outcome. The fail→pass to pass→fail asymmetry is large in every cell ( $\sim 80\times$  on 4B-with-CoT,  $\sim 44\times$  on 8B-with-CoT, with comparable ratios on the without-CoT subset), consistent with the prefix-failure mechanism in Sec. 4, i.e., reference-guided refinement primarily corrects dead-end prefixes rather than disturbing already-correct ones. Refinement is not a panacea, around a quarter to a third of  $y_o$  failures still survive after refinement and a sub-1% pass→fail leakage remains in every setting, but the 8B runs consistently lift both the recovery rate and absolute accuracy, suggesting the residual fail→fail mass shrinks as the student grows more capable of producing  $y_o$  and consuming  $y^*$ .

## B.2 Test Rollout Analysis

**Setup.** We take the same Qwen3-8B checkpoints used for Tabs. 1 and 2 and analyze their test-time rollouts on AMOBench (An et al., 2025), the most distillation-sensitive of our five benchmarks (largest absolute Pass@16 gain in Tab. 2). For each of the 39 AMOBench questions we draw  $K=16$  independent completions per method, with the same generation parameters as in Sec. 6.1 (temperature 0.6, top- $p=0.95$ , 38,912-token response budget). Each completion is then (I) tokenized with the Qwen3-8B tokenizer to obtain its output length, and (II) graded by the AMOBench rule-based verifier. Among the four dense-KL baselines we compare against +Forward KL (no clip), the strongest exploration-side baseline on AMOBench Pass@16 in Tab. 2.

**Base-difficulty buckets.** To isolate where each method helps, we partition the 39 AMOBench questions by the base model’s per-question pass count  $b_q := \sum_{i=1}^{16} \text{Verify}(y_q^{(i,\text{base})}, y_q^*) \in \{0, \dots, 16\}$  (the number of base-model rollouts that pass the verifier). This yields three difficulty buckets:

- **B0** ( $n = 23$ ): questions on which the base model fails all 16 attempts. By construction, these are *unreachable* for the base policy at  $K=16$  sampling, so any positive Pass@16 in this bucket reflects support expansion rather than sharpening.
- **B1–8** ( $n = 12$ ): medium-difficulty questions the base model solves between 1 and 8 times out of 16, where Avg@16 has the most headroom and sharpening is meaningful.
- **B9–16** ( $n = 4$ ): easy questions the base model already solves at least 9 times out of 16, near the saturation ceiling for both Avg@16 and Pass@16.

517 Bucket sizes are determined by the base model and held fixed when scoring +Forward KL and SRD,  
518 so the same question belongs to the same bucket across all three methods.

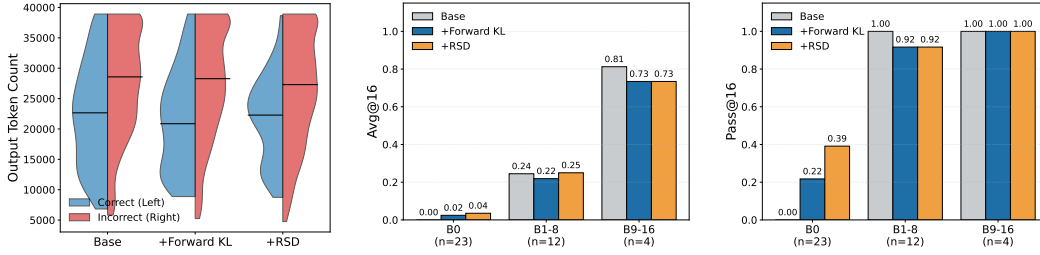


Figure 7: Test-rollout analysis on AMOBench (Qwen3-8B,  $K=16$  samples per question). **Left:** Per-rollout response-length distribution split by correctness (correct on the left half, incorrect on the right half of each violin). **Middle / Right:** Avg@16 and Pass@16 stratified by base-model difficulty bucket, where B0 groups the 23 questions the base model fails on all 16 attempts, B1–8 the 12 questions solved between 1 and 8 times, and B9–16 the 4 questions solved at least 9 times.

519 **SRD finds shorter solution paths.** The left panel of Fig. 7 shows that SRD’s correct-rollout  
520 distribution is bimodal, with a pronounced low-length mode (around  $10^4$  tokens) absent in both Base  
521 and +Forward KL, indicating that SRD finds noticeably shorter reasoning chains on the problems it  
522 can solve. Incorrect distributions are similar across methods (bunched against the generation cap),  
523 so the accuracy gains below come without longer reasoning, i.e., the (accuracy, compute) trade-off  
524 moves in the right direction.

525 **Avg@16 by bucket: sharpening on medium difficulty.** The middle panel localizes the per-sample  
526 improvement. On B1–8, SRD lifts Avg@16 to 0.25, above both Base (0.24) and +Forward KL (0.22).  
527 The fact that +Forward KL *regresses* the base model on the same bucket indicates that this sharpening  
528 is SRD-specific rather than a generic property of dense-KL distillation. On B0, the absolute Avg@16  
529 is small (0.04 for SRD vs 0.02 for +Forward KL), but every positive sample on B0 is a trajectory the  
530 base policy never produces under  $K=16$ , so SRD’s per-sample exploration rate on these questions is  
531 roughly twice that of +Forward KL.

532 **Pass@16 by bucket: frontier expansion on hard questions.** The right panel makes the exploration  
533 story explicit. On B0, the 23 AMOBench questions where the base model fails on all 16 attempts,  
534 SRD achieves Pass@16 = 0.39, nearly doubling the strongest baseline (+Forward KL at 0.22).  
535 Because B0 questions are by construction unreachable for the base policy at  $K=16$ , any positive  
536 Pass@16 here is direct evidence that SRD expands the reachable support of the base policy rather than  
537 only sharpening the existing distribution. The mild regression on B1–8 Pass@16 (–1/12 questions)  
538 is overwhelmed by the +9/23 gain on B0, leaving the dataset-level Pass@16 in Tab. 2 net positive.

### 539 B.3 Ablation: Trajectory-Subset for OPSD on Qwen3-8B

540 Each table fixes one algorithm and varies the training corpus by filtering on the outcome of  $y_o$  and  
541  $y_r$ . The three subset filters partition  $(x, y_o, y_r)$  tuples along the student’s verifier outcome on  $y_o$ . *fail*  
542 keeps tuples where  $y_o$  is incorrect ( $n=12,318$ ), *succ* keeps those where  $y_o$  is correct ( $n=27,927$ ),  
543 and *fail*→*succ* keeps the intersection where  $y_o$  is incorrect *and*  $y_r$  is correct ( $n=4,372$ ). The full  
544 corpus is the union *fail*  $\cup$  *succ* over  $n=40,245$  DeepScaleR problems and reproduces the no-subset  
545 main-table result for each algorithm. Subset rows below the rule report the change relative to the  
546 algorithm’s no-subset row (e.g., Forward-KL subset rows compare against +Forward KL, not Base).

547 **Both fail and succ halves are necessary for coverage.** Across Tabs. 3 to 5, no subset filter is  
548 uniformly better than the no-subset corpus for any algorithm. Per-column wins under a filter (e.g.,  
549 +Forward KL (fail) BeyondAIME Avg@16, +Reverse KL (fail) BeyondAIME Pass@16) are paid for  
550 by regressions elsewhere in the same row. Both halves contribute training coverage the per-token KL  
551 exploits, so the full corpus is the right default.

Table 3: Forward-KL ablation by training subset (Qwen3-8B). Red cells mark drops below that reference; bold marks the column maximum.

| Method             | Traj. | Avg@16 (%)  |             |             |                    |             | Pass@16 (%) |                    |                    |             |             |
|--------------------|-------|-------------|-------------|-------------|--------------------|-------------|-------------|--------------------|--------------------|-------------|-------------|
|                    |       | AIME24      | AIME25      | HMMT25      | BeyondAIME         | AMOBench    | AIME24      | AIME25             | HMMT25             | BeyondAIME  | AMOBench    |
| Base               | –     | 76.5        | 66.7        | 41.5        | 41.6               | 15.9        | 90.0        | 83.3               | 66.7               | 66.0        | 41.0        |
| +SRD (ours)        | $y_r$ | <b>76.5</b> | <b>69.2</b> | <b>44.5</b> | <b>42.8</b>        | <b>17.3</b> | <b>90.0</b> | <b>86.7</b>        | <b>73.3</b>        | <b>68.0</b> | <b>61.5</b> |
| +Forward KL        | $y_o$ | 74.8        | 65.4        | 40.3        | 39.6               | 15.7        | 86.7        | 82.7               | <b>73.3</b>        | <b>68.0</b> | 51.3        |
| +Forward KL (fail) | $y_o$ | 74.9 (+0.1) | 68.1 (+2.7) | 40.1 (-0.2) | <b>42.8</b> (+3.2) | 16.3 (+0.6) | 85.3 (-1.4) | <b>86.7</b> (+4.0) | 70.0 (-3.3)        | 67.0 (-1.0) | 46.2 (-5.1) |
| +Forward KL (succ) | $y_o$ | 75.0 (+0.2) | 67.7 (+2.3) | 40.1 (-0.2) | 40.1 (+0.5)        | 15.8 (+0.1) | 86.7 (+0.0) | 83.3 (+0.6)        | <b>73.3</b> (+0.0) | 64.6 (-3.4) | 46.2 (-5.1) |

Table 4: Reverse-KL ablation by training subset (Qwen3-8B). Red cells mark drops below that reference; bold marks the column maximum.

| Method             | Traj. | Avg@16 (%)  |                    |             |             |             | Pass@16 (%) |                    |             |                    |             |
|--------------------|-------|-------------|--------------------|-------------|-------------|-------------|-------------|--------------------|-------------|--------------------|-------------|
|                    |       | AIME24      | AIME25             | HMMT25      | BeyondAIME  | AMOBench    | AIME24      | AIME25             | HMMT25      | BeyondAIME         | AMOBench    |
| Base               | –     | 76.5        | 66.7               | 41.5        | 41.6        | 15.9        | 90.0        | 83.3               | 66.7        | 66.0               | 41.0        |
| +SRD (ours)        | $y_r$ | <b>76.5</b> | <b>69.2</b>        | <b>44.5</b> | <b>42.8</b> | <b>17.3</b> | <b>90.0</b> | <b>86.7</b>        | <b>73.3</b> | 68.0               | <b>61.5</b> |
| +Reverse KL        | $y_o$ | 75.4        | 68.2               | 44.3        | 40.8        | 16.8        | 86.7        | <b>86.7</b>        | <b>73.3</b> | 64.0               | 51.3        |
| +Reverse KL (fail) | $y_o$ | 75.4 (+0.0) | <b>69.2</b> (+1.0) | 44.0 (-0.3) | 42.4 (+1.6) | 16.7 (-0.1) | 83.3 (-3.4) | <b>86.7</b> (+0.0) | 66.7 (-6.6) | <b>70.0</b> (+6.0) | 53.8 (+2.5) |
| +Reverse KL (succ) | $y_o$ | 75.4 (+0.0) | <b>69.2</b> (+1.0) | 44.2 (-0.1) | 42.3 (+1.5) | 16.2 (-0.6) | 83.3 (-3.4) | <b>86.7</b> (+0.0) | 70.0 (-3.3) | 68.0 (+4.0)        | 46.2 (-5.1) |

Table 5: SRD ablation by training subset (Qwen3-8B). Red cells mark drops below that reference; bold marks the column maximum.

| Method                           | Traj. | Avg@16 (%)  |                    |             |                    |                    | Pass@16 (%) |                    |             |             |              |
|----------------------------------|-------|-------------|--------------------|-------------|--------------------|--------------------|-------------|--------------------|-------------|-------------|--------------|
|                                  |       | AIME24      | AIME25             | HMMT25      | BeyondAIME         | AMOBench           | AIME24      | AIME25             | HMMT25      | BeyondAIME  | AMOBench     |
| Base                             | –     | 76.5        | 66.7               | 41.5        | 41.6               | 15.9               | 90.0        | 83.3               | 66.7        | 66.0        | 41.0         |
| +SRD (ours)                      | $y_r$ | <b>76.5</b> | 69.2               | <b>44.5</b> | 42.8               | <b>17.3</b>        | <b>90.0</b> | 86.7               | <b>73.3</b> | <b>68.0</b> | <b>61.5</b>  |
| +SRD (fail $\rightarrow \cdot$ ) | $y_r$ | 75.4 (-1.1) | 69.2 (+0.0)        | 41.9 (-2.6) | <b>43.3</b> (+0.5) | <b>17.3</b> (+0.0) | 86.7 (-3.3) | 83.3 (-3.4)        | 66.7 (-6.6) | 64.0 (-4.0) | 51.3 (-10.2) |
| +SRD (succ $\rightarrow \cdot$ ) | $y_r$ | 75.4 (-1.1) | <b>70.4</b> (+1.2) | 43.5 (-1.0) | <b>43.3</b> (+0.5) | 15.9 (-1.4)        | 86.7 (-3.3) | <b>86.9</b> (+0.2) | 70.0 (-3.3) | 65.0 (-3.0) | 46.2 (-15.3) |
| +SRD (fail $\rightarrow$ succ)   | $y_r$ | 75.4 (-1.1) | 69.0 (-0.2)        | 42.9 (-1.6) | 43.1 (+0.3)        | 16.8 (-0.5)        | 86.7 (-3.3) | 83.3 (-3.4)        | 70.0 (-3.3) | 67.0 (-1.0) | 51.3 (-10.2) |

**Filtering is not an optimization of Eq. (6).** Subset filtering changes which trajectories enter  $\mathcal{D}$  but not the loss itself, so neither *succ*-only nor *fail*-only is an optimization of Eq. (6). Each filter also drops a complementary signal. *succ*-only loses the hard problems on which the student fails unaided, where the teacher provides extra signal that raises the probability of reaching previously unreachable solutions. *fail*-only loses the alternative-path signal on the easy half, where the teacher can offer stronger or shorter derivations than the student would produce on its own.

**Forward KL is the most data-sensitive.** Vanilla +Forward KL regresses Base on four of five Avg@16 benchmarks (AIME24  $-1.7$ , AIME25  $-1.3$ , HMMT25  $-1.2$ , BeyondAIME  $-2.0$ ). Filtering recovers AIME25 (fail  $+2.7$ , succ  $+2.3$ ) and BeyondAIME (fail  $+3.2$ ), but AIME24 and HMMT25 remain below Base under any filter, and Pass@16 is mostly traded down (AMOBench  $-5.1$  under both filters). This column-specific trade is consistent with the mass-spread character of the mode-covering forward KL, which cannot ignore points in the corpus and therefore inherits both the support coverage and the prefix-failure pressure of whichever subset it is fed.

**Reverse KL is almost flat under filtering.** Both subset rows sit within  $\pm 1.0$  of the no-filter Avg@16 and trade Pass mildly (BeyondAIME up, AMOBench down). The mode-seeking reverse KL already discounts low-probability regions, so removing the fail or succ half does not change the optimization much. This is a stability story, not a quality story, and the no-subset Reverse KL is therefore not noticeably improved by curation.

**SRD is hurt by filtering, especially on Pass@16.** Every subset row drops Pass@16 on every benchmark, with double-digit losses on AMOBench ( $-10$  to  $-15$ ). Avg@16 deltas are small and mixed. The fail $\rightarrow$ succ row ( $n=4,372$ , the "ideal" subset where refinement fixed errors) does not outperform full corpus, exhibiting the same Pass losses and no Avg gains worth the data cut. SRD's gain comes from the breadth of the refined corpus rather than from any privileged subset, supporting the choice to train on  $y_r$  over the unfiltered corpus by default.

## C Experiment Details

This appendix collects the training data and trajectory construction (Sec. C.1), hardware and wall-clock budget (Sec. C.3), the refinement prompt template (Sec. C.4), the training-metric extraction used in Fig. 2 (Sec. C.5), models and distillation setup (Sec. C.6), method-specific and common optimization hyperparameters (Secs. C.7 and C.8), and the evaluation protocol (Sec. C.2) used throughout Sec. 6.

### C.1 Training Data and Trajectory Construction

We train on DeepScaleR (Luo et al., 2025), a math problem corpus with verified answers, approximately 40K problems each with a reference solution  $y^*$ . The release is de-duplicated against AIME problems, ensuring no overlap with AIME24 in our evaluation; AIME25, HMMT25, BeyondAIME, and AMOBench post-date the corpus and are by construction non-overlapping. For every problem  $x$  we run a two-stage offline pipeline before any KL update.

**Stage 1 (raw rollout  $y_o$ ).** The current student samples one rollout per prompt with vLLM (?) using temperature 0.6, top- $p=0.95$ , top- $k=20$ , prompt budget 4096 tokens, and response budget 16,384 tokens. Each rollout is scored by a rule-based verifier (DeepSeek-AI, 2025) against  $y^*$  to populate a binary outcome label used only for the  $D$  bucketing in Fig. 2 (training itself does not use this label).

**Stage 2 (refined trajectory  $y_r$ , SRD only).** Conditioning on the same problem, the raw rollout, and the reference, the student samples  $y_r \sim \pi_\theta(\cdot \mid x, y_o, y^*)$  via the prompt template in Sec. C.4. Sampling parameters match Stage 1 except the prompt budget grows to 24,576 tokens to accommodate the concatenated initial-solution and reference fields. Stage 2 runs over all problems, regardless of whether  $y_o$  passed Stage 1. The dense-KL baselines train on the Stage 1 parquet and skip Stage 2.

### C.2 Evaluation Protocol

Evaluation is run on the merged-weights checkpoint after Stage 2 training. We sample 16 completions per problem with vLLM at temperature 0.6, top- $p=0.95$ , prompt budget 4,096 tokens, and response budget 38,912 tokens (chosen to avoid truncation on long thinking traces). Each completion is scored by the same rule-based verifier used in Stage 1, with answer extraction from the final `\boxed{...}` block.

**Metrics.** For each test question  $x_q$  we draw  $K=16$  independent completions  $y_q^{(i)} \sim \pi_\theta(\cdot \mid x_q)$  and score them with the verifier  $\text{Verify}(\cdot, \cdot)$  from Eq. (5). Over  $N$  test questions,

$$\text{Avg@}K = \frac{1}{N} \sum_{q=1}^N \frac{1}{K} \sum_{i=1}^K \text{Verify}(y_q^{(i)}, y_q^*), \quad \text{Pass@}K = \frac{1}{N} \sum_{q=1}^N \max_{1 \leq i \leq K} \text{Verify}(y_q^{(i)}, y_q^*).$$

Avg@16 tracks sharpening of the modal answer; Pass@16 tracks the breadth of the student’s reasoning support.

**Benchmarks.** AIME24 and AIME25 (American Invitational Mathematics Examination, 30 problems each) (Mathematical Association of America, 2024, 2025); HMMT25 (Harvard-MIT Mathematics Tournament February 2025, 30 problems) (HMMT, 2025); BeyondAIME (a curated set of competition problems harder than AIME) (Seed, 2025); and AMOBench (Asian Mathematical Olympiad benchmark) (An et al., 2025). AMOBench consists of 50 hand-designed problems, 39 with parser-based grading and 11 with descriptive (free-form) answers; we evaluate on the 39 parser-graded problems only, so they share the same rule-based grader as the other four benchmarks. All numbers reported under *AMOBench* therefore refer to this 39-problem subset.

### C.3 Hardware and Compute

All runs use a single node of  $8 \times$  H100 80GB GPUs with FSDP2 sharding via verl (Sheng et al., 2024). Each row of Tabs. 1 and 2 corresponds to one offline pipeline run, comprising Stage 1 generation (raw rollout  $y_o$ ), optional Stage 2 generation (refined trajectory  $y_r$ , SRD only), and one training epoch over the resulting parquet.

**Training-time trade-off.** A non-obvious consequence of the trajectory compression in [Sec. B.1](#) is a substantial training-time saving. Dense-KL baselines train on  $y_o$  (median  $\sim 7.7\text{K}$  tokens on Qwen3-8B); SRD trains on  $y_r$  (median  $\sim 0.88\text{K}$ ). Per-step compute scales with two length-dependent quantities, (I) transformer forward/backward (attention quadratic, MLP linear in length), and (II) the full-vocabulary KL head, which materializes  $|\mathcal{V}| \approx 152\text{K}$  logits per kept token. The  $\sim 9\times$  shorter  $y_r$  therefore translates almost directly into faster training even at matched effective batch.

**Measured wall-clock.** [Tab. 6](#) reports per-method end-to-end wall-clock, organized by the stages each method actually runs. Forward KL and Reverse KL train on  $y_o$ , so they only pay the Stage 1 sampling cost before training. SRD additionally runs Stage 2 (sampling  $y_r$  via the refinement prompt of [Sec. C.4](#)), but its training step operates on the much shorter  $y_r$  and finishes in roughly 40% of the time, e.g., 2h10m on Qwen3-8B vs 5h30m for Reverse KL on  $y_o$ . On Qwen3-8B the training saving more than cancels the Stage 2 cost, so SRD runs end-to-end in 9h20m, 20–30 minutes faster than the dense-KL baselines (9h40m / 9h50m) despite paying for an additional sampling stage. On Qwen3-4B the absolute training time is small to begin with, so SRD pays a residual  $\sim 1\text{h}10\text{m}$  overhead, but the absolute scale is short.

Table 6: End-to-end wall-clock per method (h:mm, rounded to 10 minutes) on a single  $8\times \text{H100}$  80GB node. Forward KL and Reverse KL train on  $y_o$  and skip Stage 2; SRD additionally runs Stage 2 but trains on the much shorter  $y_r$ . Evaluation ([Tab. 7](#)) is shared across methods and excluded from the total.

| Backbone               | Method     | Stage 1 ( $y_o$ ) | Stage 2 ( $y_r$ ) | Training | Total |
|------------------------|------------|-------------------|-------------------|----------|-------|
| Qwen3-4B-Instruct-2507 | Forward KL | 2:10              | —                 | 2:10     | 4:20  |
|                        | Reverse KL | 2:10              | —                 | 2:10     | 4:20  |
|                        | SRD (ours) | 2:10              | 2:00              | 1:20     | 5:30  |
| Qwen3-8B               | Forward KL | 4:20              | —                 | 5:20     | 9:40  |
|                        | Reverse KL | 4:20              | —                 | 5:30     | 9:50  |
|                        | SRD (ours) | 4:20              | 2:50              | 2:10     | 9:20  |

We therefore view the Stage 2 sampling pass not as a strict additional cost but as a budget redistribution from training to inference, with a favorable signal-quality trade ([Sec. B.1](#)).

**Evaluation wall-clock.** [Tab. 7](#) reports per-benchmark generation time for the  $K=16$  rollouts per question used to compute Avg@16 and Pass@16. The same evaluation budget is paid by every method (Base, dense-KL baselines, SRD), so it does not affect the relative comparison in [Tab. 6](#).

Table 7: Evaluation wall-clock per benchmark (h:mm, rounded to 10 minutes) on a single  $8\times \text{H100}$  80GB node. *Questions* is the original test-set size; *Rollouts* =  $N \times K$  with  $K=16$  samples per question.

| Benchmark  | Questions | Rollouts | 4B time     | 8B time     |
|------------|-----------|----------|-------------|-------------|
| AIME24     | 30        | 480      | 0:20        | 0:30        |
| AIME25     | 30        | 480      | 0:20        | 0:40        |
| HMMT25     | 30        | 480      | 0:20        | 0:30        |
| BeyondAIME | 100       | 1,600    | 0:30        | 1:20        |
| AMOBench   | 39        | 624      | 0:20        | 1:00        |
| Total      | 229       | 3,664    | $\sim 2:00$ | $\sim 4:00$ |

#### C.4 Refinement Prompt

For each problem  $x$ , raw rollout  $y_o$ , and reference solution  $y^*$ , we use the following instruction to sample the refined trajectory  $y_r$ :



**SRD Prompt**

Your task is to rewrite your mathematical solution using the reference solution as guidance.

**Problem:**

{PROBLEM}

**Reference Solution:**

{EXPERT SOLUTION}

**Your Initial Solution:**

{INITIAL RESPONSE}

**Instructions:**

1. Review the reference solution to understand the target reasoning and method
2. Rewrite your solution so it is consistent with the reference solution
3. Keep useful parts of original solutions' structure and style when appropriate
4. Output ONLY the rewritten solution

## 644 C.5 Training Metrics for Fig. 2

645 We log three diagnostics during OPSD training, all computed on student rollouts  $y_o \sim \pi_\theta(\cdot | x)$  over  
 646 response tokens (with mask  $m_{i,t} \in \{0, 1\}$ ). All three are first accumulated as numerator/denominator  
 647 within each step, then divided, so the reported value is a token-weighted mean.

648 **Per-token KL by rollout outcome.** Each prompt  $x_i$  carries a binary stage-1 outcome label  $b_i$ , set  
 649 to 1 (*correct*) if the base model's stage-1 rollout passes the verifier and to 0 (*incorrect*) otherwise.  
 650 The per-token KL between teacher and student is averaged separately within each bucket,

$$D^{\text{correct}} = \frac{\sum_{i: b_i=1} \sum_t D_{\text{KL},i,t} m_{i,t}}{\sum_{i: b_i=1} \sum_t m_{i,t}}, \quad D^{\text{incorrect}} = \frac{\sum_{i: b_i=0} \sum_t D_{\text{KL},i,t} m_{i,t}}{\sum_{i: b_i=0} \sum_t m_{i,t}},$$

651 where  $D_{\text{KL},i,t}$  is the per-token KL term in Eq. (1). The values are directly comparable to the global  
 652  $D$  since both use a token-weighted denominator.

653 **Epistemic-token mass.** Let  $\mathcal{E} \subset \mathcal{V}$  be the set of *epistemic onset* tokens. We construct  $\mathcal{E}$  from 16  
 654 phrases,

655 Wait, Actually, However, Alternatively, Oops, Wrong, Error, Incorrect, Correction,  
 656 Sorry, Hmm, Oh, Hold, Pause, Uh, Um,

657 by tokenising each phrase under the student tokenizer in two variants (bare string and leading-space)  
 658 and collecting the first sub-word id, then deduplicating. At each response position  $(i, t)$  we measure  
 659 how much teacher mass falls on  $\mathcal{E}$  before any KL temperature scaling,

$$\text{mass}_{i,t} = \sum_{v \in \mathcal{E}} \pi_T(v | x_i, y_i^*, y_{<t}) = \sum_{v \in \mathcal{E}} \text{softmax}(\ell_{i,t}^T)_v,$$

660 and report the token-weighted mean  $(\sum_{i,t} \text{mass}_{i,t} m_{i,t}) / (\sum_{i,t} m_{i,t})$ . The metric is computed only  
 661 on the full-vocabulary path (i.e., when teacher logits are materialised) and is independent of the  
 662 KL-loss temperature.

663 **Teacher-student perplexity gap.** Both perplexities are token-weighted under teacher-forced de-  
 664 coding on the same response mask,

$$\text{PPL}_S = \exp\left(\frac{\sum_{i,t} -\log \pi_S(y_{i,t} | y_{<t}) m_{i,t}}{\sum_{i,t} m_{i,t}}\right), \quad \text{PPL}_T = \exp\left(\frac{\sum_{i,t} -\log \pi_T(y_{i,t} | y_{<t}) m_{i,t}}{\sum_{i,t} m_{i,t}}\right).$$

665 Since both share the same mask and use  $T=1$  log-probabilities, the gap  $\text{PPL}_S - \text{PPL}_T$  is directly  
 666 comparable across runs.

## C.6 Models and Distillation Setup

Base models are Qwen3-4B-Instruct-2507 and Qwen3-8B with thinking enabled (Yang et al., 2025). Throughout, the same backbone embodies teacher and student via the privileged-context substitution  $\pi_T(\cdot \mid x, y_{<t}) = \pi_\theta(\cdot \mid x, y^*, y_{<t})$  (Eq. (1)); LoRA adapters update only the student branch, and the stop-gradient operator on the teacher branch is implemented by detaching  $\pi_T$ ’s logits before the KL is computed. We apply LoRA of rank  $r=64$ , scaling  $\alpha=128$ , dropout 0.05, on all attention and MLP linear layers (q\_proj, k\_proj, v\_proj, o\_proj, gate\_proj, up\_proj, down\_proj). After training we merge the adapters into the base weights for evaluation.

## C.7 Method-Specific Hyperparameters

Tab. 8 lists the per-method settings that distinguish each row of Tabs. 1 and 2. All five methods use the full-vocabulary KL estimator (loss head materialized over the full Qwen3 vocabulary,  $|\mathcal{V}| \approx 152\text{K}$ ), AdamW, bfloat16, gradient checkpointing, and a single training epoch. Effective batch size is computed as (per-GPU batch)  $\times$  (GPUs)  $\times$  (gradient accumulation), assuming 8 GPUs per node.

Table 8: Per-method training settings. *Forward KL w/ Clip* reproduces the OPSD recipe of Zhao et al. (2026); *Reverse KL w/ Top-K* reproduces the truncated-support objective of Fu et al. (2026).

| Method                    | Traj. | KL dir. | Clip $c$ | Top- $K$ | Temp. $T$ | LR                 | Eff. batch | Max len. |
|---------------------------|-------|---------|----------|----------|-----------|--------------------|------------|----------|
| Forward KL                | $y_o$ | forward | 0        | —        | 1.0       | $5 \times 10^{-6}$ | 32         | 18432    |
| Forward KL w/ Clip (OPSD) | $y_o$ | forward | 0.06     | —        | 1.1       | $5 \times 10^{-6}$ | 32         | 18432    |
| Reverse KL                | $y_o$ | reverse | 0        | —        | 1.0       | $2 \times 10^{-5}$ | 256        | 18432    |
| Reverse KL w/ Top- $K$    | $y_o$ | reverse | 0        | 32       | 1.0       | $2 \times 10^{-6}$ | 128        | 18432    |
| SRD (ours)                | $y_r$ | forward | 0        | —        | 1.0       | $2 \times 10^{-5}$ | 512        | 20480    |

## C.8 Common Optimization Hyperparameters

Settings shared across all methods are listed in Tab. 9. Reverse KL w/ Top- $K$  uses zero warmup and zero weight decay following Fu et al. (2026); all other methods use the values in the table.

Table 9: Common optimization hyperparameters.

| Setting                | Value                                                          |
|------------------------|----------------------------------------------------------------|
| Optimizer              | AdamW ( $\beta_1=0.9$ , $\beta_2=0.999$ , $\epsilon=10^{-8}$ ) |
| Precision              | bfloat16                                                       |
| Gradient checkpointing | enabled                                                        |
| Gradient clipping      | 1.0                                                            |
| LR schedule            | linear warmup, cosine decay to $0.1 \times$ peak LR            |
| Warmup ratio           | 0.1 (Reverse KL w/ Top- $K$ uses 0)                            |
| Weight decay           | 0.0 (Reverse KL uses 0.005)                                    |
| Epochs                 | 1                                                              |
| Sequence packing       | remove-padding via verl FSDP2                                  |
| Sequence parallel      | ulysses, size 1                                                |

**Loss formulation.** For each method, the per-token KL is computed in full vocabulary at temperature  $T$  as

$$D^{(T)}(p \parallel q) = \sum_{v \in \mathcal{V}} p_T(v) (\log p_T(v) - \log q_T(v)),$$

with  $p_T(v) \propto \exp(\ell_v/T)$ . The clipping baseline applies the OPSD per-token cap  $\min(D_{\text{KL},t}, c)$  before averaging over the response mask. Top- $K$  replaces  $\mathcal{V}$  by the teacher’s top-32 support  $\mathcal{S}_t$  and renormalizes both  $p_T$  and  $q_T$  to sum to 1 on  $\mathcal{S}_t$  before evaluating  $D$ .

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| 1023 | Question: Does the paper describe the usage of LLMs if it is an important, original, or non-standard component of the core methods in this research? Note that if the LLM is used only for writing, editing, or formatting purposes and does <i>not</i> impact the core methodology, scientific rigor, or originality of the research, declaration is not required. |
| 1024 |                                                                                                                                                                                                                                                                                                                                                                     |
| 1025 |                                                                                                                                                                                                                                                                                                                                                                     |
| 1026 |                                                                                                                                                                                                                                                                                                                                                                     |
| 1027 | Answer: [N/A]                                                                                                                                                                                                                                                                                                                                                       |
| 1028 | Justification: LLMs are the object of study (the student/teacher being distilled, Qwen3-4B and Qwen3-8B), not an external non-standard component of the methodology; their use is fully described in <a href="#">Sec. 6.1</a> and <a href="#">Sec. C</a> .                                                                                                          |
| 1029 |                                                                                                                                                                                                                                                                                                                                                                     |
| 1030 |                                                                                                                                                                                                                                                                                                                                                                     |
| 1031 | Guidelines:                                                                                                                                                                                                                                                                                                                                                         |
| 1032 | <ul style="list-style-type: none"> <li>• The answer [N/A] means that the core method development in this research does not involve LLMs as any important, original, or non-standard components.</li> </ul>                                                                                                                                                          |
| 1033 |                                                                                                                                                                                                                                                                                                                                                                     |
| 1034 | <ul style="list-style-type: none"> <li>• Please refer to our LLM policy in the NeurIPS handbook for what should or should not be described.</li> </ul>                                                                                                                                                                                                              |
| 1035 |                                                                                                                                                                                                                                                                                                                                                                     |